香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen

SCHOOL OF DATA SCIENCE
數據科學學院

CSC 6011 - Theory of Computation

Lecture 18 - L and NL

Time: Mon & Wed, 4:00 - 5:20 PM

Location: Teaching A, 107

Instructor: Tao LIN (林涛)

TA: Yefan GAO (高叶繁)

Fall 2026

Two proposals

Vote YES/NO:

- Final exam will contain some questions (such as Mini-Quiz) in the slides.
- Answers to those questions will be shown in class, but not in the slides uploaded after class.

Lecture 18 - L and NL

Last time:

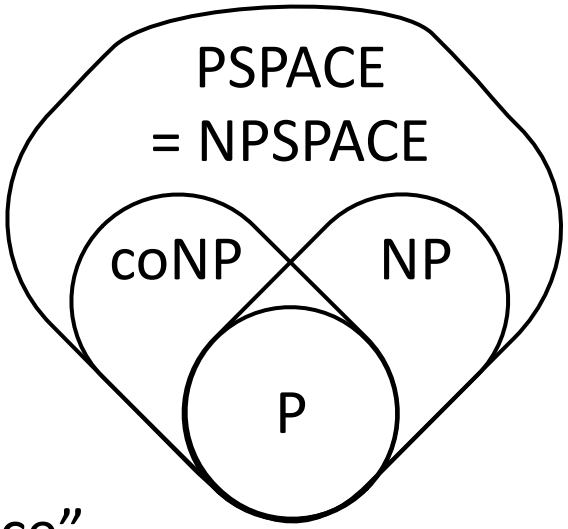
- $\mathrm{LADDER}_{\mathrm{DFA}} \in$ PSPACE
- Savitch's Theorem: $\mathrm{NSPACE}\big(f(n)\big) \subseteq \mathrm{SPACE}\big(f^2(n)\big)$

  So, NPSPACE = PSPACE
- TQBF is PSPACE-complete

Today (Sipser §8.4 - 8.5):

- Logspace: L and NL
- L ⊆ NL ⊆ P
- NL-completeness

Review: SPACE Complexity



So far, we have:

- Been studying TMs M that run in at least linear space $f(n) \geq n$
- Defined:
  - PSPACE $= \bigcup_k \mathrm{SPACE}(n^k)$ "polynomial space"
  - NPSPACE $= \bigcup_k \mathrm{NSPACE}(n^k)$ "nondeterministic polynomial space"
- Proved the Savitch's Theorem:

$$\mathrm{NSPACE}\big(f(n)\big) \subseteq \mathrm{SPACE}\big(f^2(n)\big)$$

- Saw a connection between computation and reachability of configurations:

$$M \text{ accepts } w \iff c_{\text{start}} \text{ can reach } c_{\text{accept}}$$

We now study logarithmic space complexity:

- The entire Internet is huge -- but a URL is short.
- A file might be large -- but a pointer to a position is short.
- For n input positions, a pointer needs only $\lceil \log_2 n \rceil$ bits.

Log space

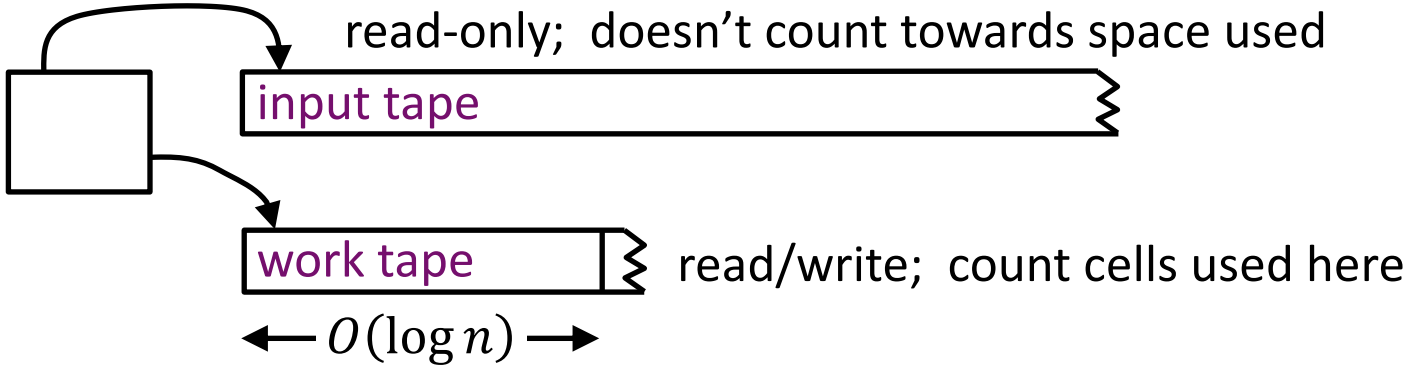
To define sublinear space complexity, do not count the input part as space used.

Use 2-tape TM model with read-only input tape.

Defn: L = $\mathrm{SPACE}(\log n)$

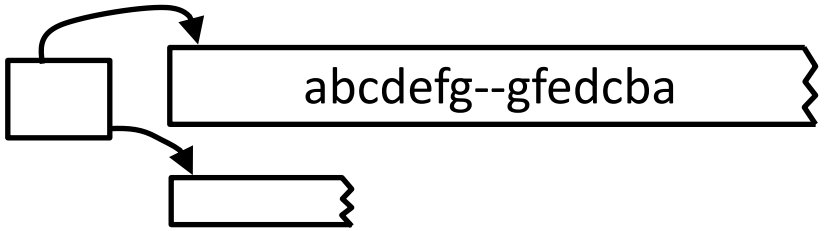
NL = $\mathrm{NSPACE}(\log n)$

Log space can represent a constant number of pointers into the input.



Examples:

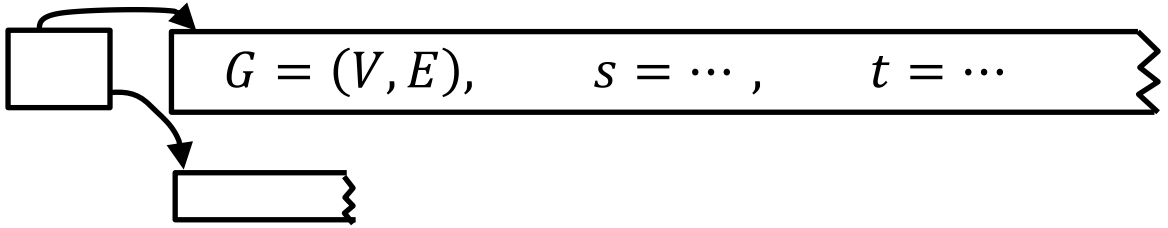
1. $\{ww^{\mathcal{R}} \mid w \in \Sigma^*\} \in$ L



Work tape tracks 2 locations on the input tape.

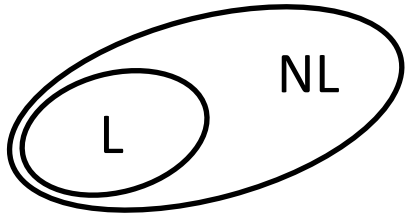
2. PATH ∈ NL

   Starting from s, nondeterministically guess the next node to go. (don't guess the entire sequence)



Work tape tracks the current node and the number of steps (stop if exceeding $|V|$).

Log space properties



Theorem: L ⊆ NL

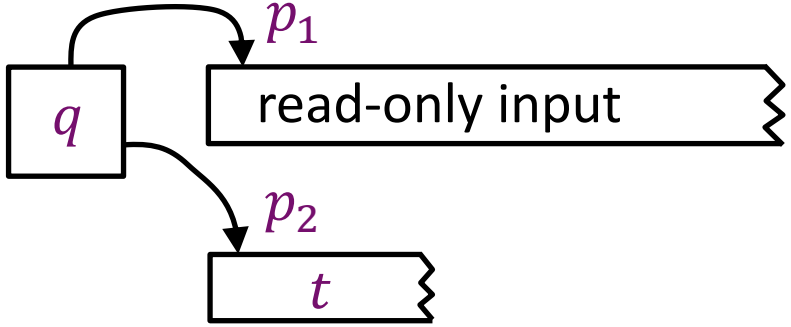
Theorem: NL ⊆ $\mathrm{SPACE}(\log^2 n)$

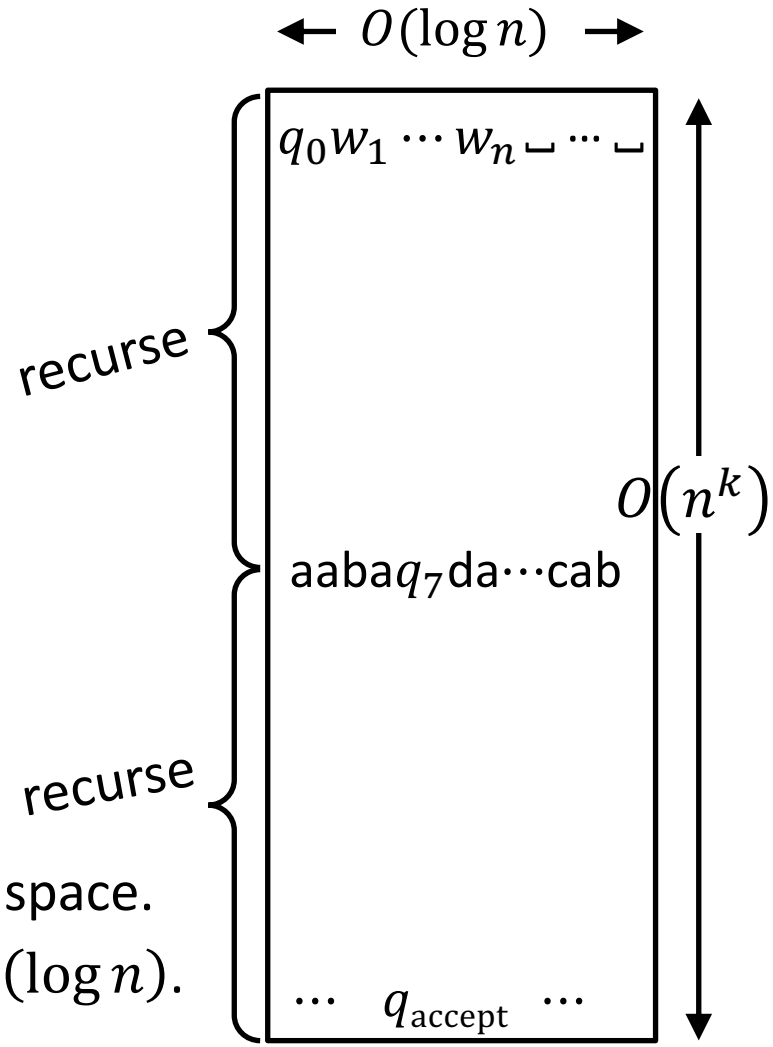
Note: this doesn't imply NL ⊆ L, because L = $\mathrm{SPACE}(\log n)$

Proof: Use Savitch's theorem: $\mathrm{NSPACE}\big(f(n)\big) \subseteq \mathrm{SPACE}\big(f^2(n)\big)$

Defn: A configuration of a 2-tape TM is (q, p_1, p_2, t) where q is a state, p_1 and p_2 are the tape head positions, and t is the work tape contents.

- The number of such configurations is
  $|Q| \times n \times O(\log n) \times d^{O(\log n)} = O(n^k)$ for some k.

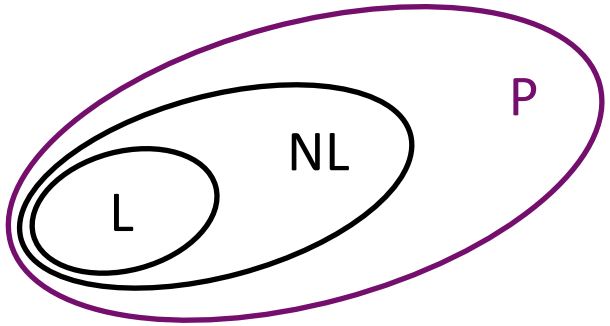




- Each recursion level uses $O(\log n)$ space.
- Number of levels = $\log O\big(n^k\big) = O(\log n)$.

Total: $O(\log^2 n)$ space.

Log space properties



Theorem: L ⊆ NL

Theorem: NL ⊆ $\mathrm{SPACE}(\log^2 n)$

Theorem: NL ⊆ P

Question 18.1:

We just proved NL ⊆ $\mathrm{SPACE}(\log^2 n)$.

We learned from Lecture 15 that $\mathrm{SPACE}(f(n)) \subseteq \mathrm{TIME}(2^{O(f(n))})$.

Can we prove NL ⊆ P by letting $f(n) = \log^2 n$?

(a) Yes (b) No

NL ⊆ P

Theorem: NL ⊆ P

Proof: Pick $A \in$ NL. An NTM M decides A in space $O(\log n)$.

Defn: The configuration graph $G_{M,w}$ for M on w has

- **nodes:** all configurations for M on w -- $O(n^k)$ many
- **edges:** edge from $c_i \to c_j$ if c_i can go to c_j in 1 step.

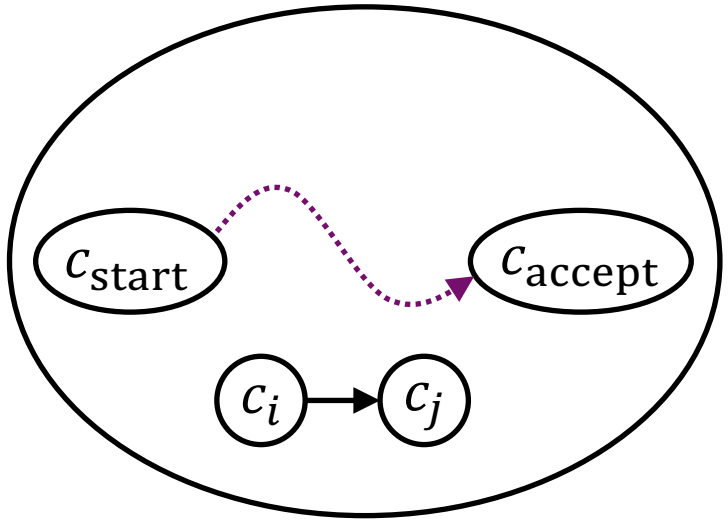
Claim: M accepts w iff the configuration graph $G_{M,w}$ has a path from c_{start} to c_{accept}

Polynomial time algorithm T for A:

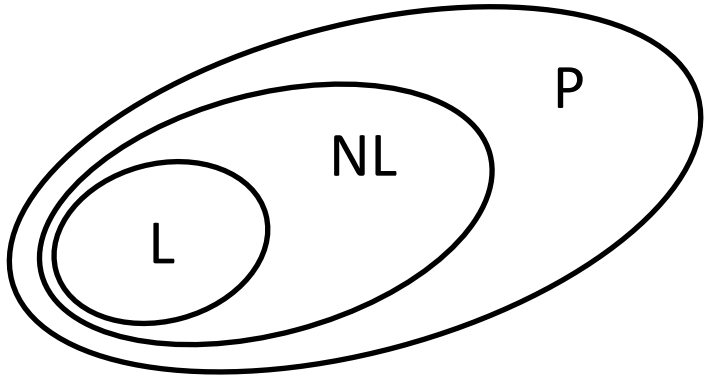
$T =$ "On input w

1. Construct $G_{M,w}$. [polynomial size]
2. *Accept* if there is a path from c_{start} to c_{accept}. *Reject* if not."

configuration graph $G_{M,w}$



iff M accepts w



L = NL? Unsolved L = P? Unsolved

NL-completeness

Defn: B is NL-complete if

1) $B \in$ NL
2) For any $A \in$ NL, $A \leq_m^L B$

Log-space reduction

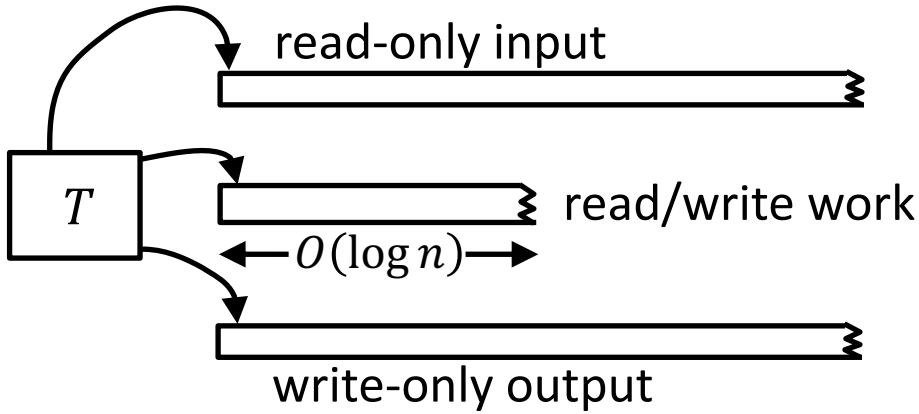
Defn: A log-space transducer is a TM with three tapes:

1. read-only input tape of size n
2. read/write work tape of size $O(\log n)$
3. write-only output tape

A log-space transducer T computes a function $f: \Sigma^* \to \Sigma^*$ if, on any input w, T halts with $f(w)$ on its output tape.

Say that f is computable in log-space.

Defn: A is log-space reducible to B ($A \leq_m^L B$) if $A \leq_m B$ by a reduction function that is computable in log-space.



Theorem: If $A \leq_m^L B$ and $B \in$ L, then $A \in$ L

Proof: TM for $A =$ "On input w

1. Compute $f(w)$ (using T)
2. Run decider for B on $f(w)$.
   Output same."

BUT, we don't have space to store $f(w)$.

So, (re-)compute $f(w)$ as needed:

- when the decider for B needs to read the i-th symbol of $f(w)$, re-run T to get the i-th symbol of $f(w)$

Mini-Quiz 18.2

If T is a log-space transducer that computes f, then for input w of length n, how long can $f(w)$ be?

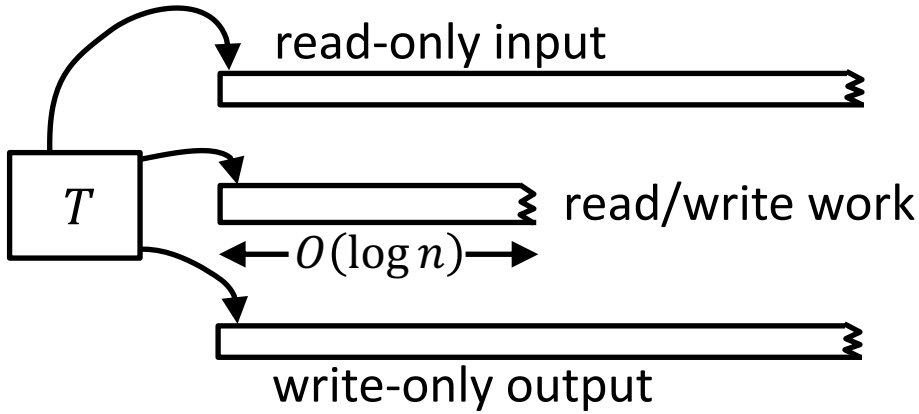
(a) at most $O(\log n)$

(b) at most $O(n)$

(c) at most polynomial in n

(d) at most $2^{O(n)}$

(e) any length



Defn: A log-space transducer is a TM with three tapes:

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A log-space transducer T computes a function $f: \Sigma^* \to \Sigma^*$ if, on any input w, T halts with $f(w)$ on its output tape.

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BUT, we don't have space to store $f(w)$.

So, (re-)compute $f(w)$ as needed:

- when the decider for B needs to read the i-th symbol of $f(w)$, re-run T to get the i-th symbol of $f(w)$

PATH is NL-complete

Theorem: PATH is NL-complete

Proof: 1) PATH ∈ NL ✓

2) For any $A \in$ NL, $A \leq_m^L$ PATH

Let $A \in$ NL be decided by NTM M in $O(\log n)$ space.

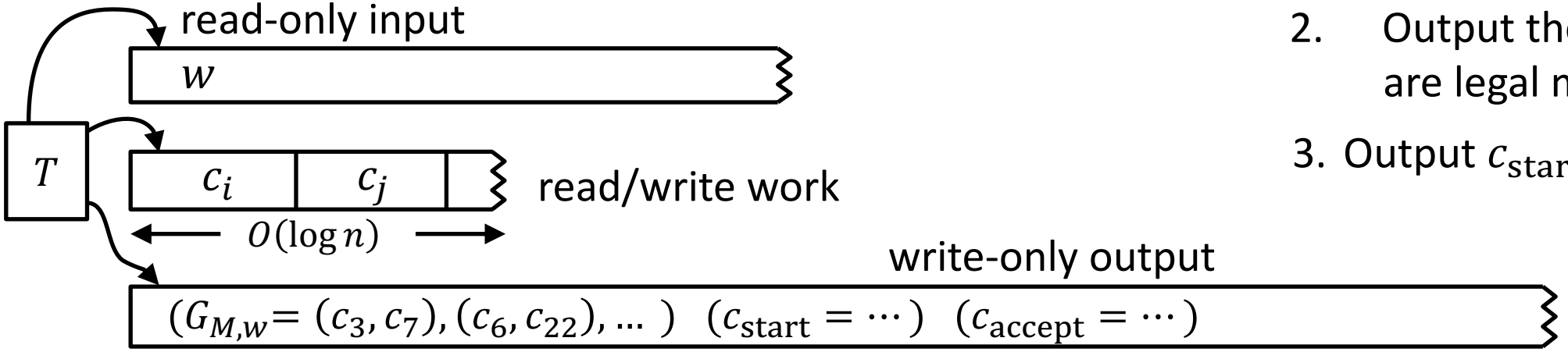
[Assume that M has a single accepting configuration.]

Give a log-space reduction f mapping A to PATH.

$f(w) = \langle G, s, t \rangle = \langle G_{M,w}, c_{\text{start}}, c_{\text{accept}} \rangle$

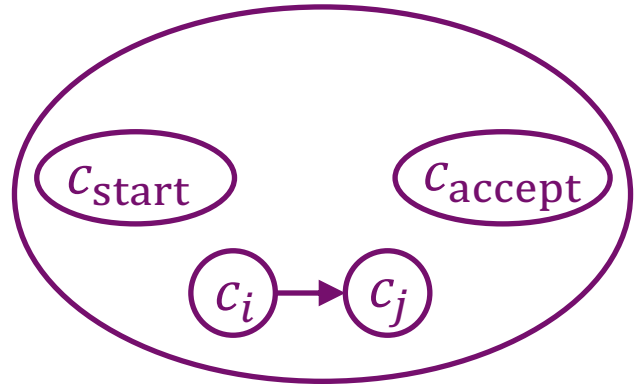
$w \in A$ iff G has a path from s to t

A log-space transducer T to compute f in log-space:



The configuration graph $G_{M,w}$

$f(w) =$



$T =$ "On input w

1. For all pairs c_i, c_j of configurations of M on w.
2. Output those pairs that are legal moves of M.
3. Output c_{start} and c_{accept}."

$\overline{\text{2SAT}}$ is NL-complete

Example of 2CNF: (See homework 3)

$$\varphi = (x_1 \vee \overline{x_2}) \wedge (\overline{x_1} \vee x_3) \wedge (x_2 \vee \overline{x_3})$$

$$\overline{\text{2SAT}} = \{\langle\varphi\rangle \mid \varphi \text{ is an unsatisfiable 2CNF formula}\}$$

Theorem: $\overline{\text{2SAT}}$ is NL-complete

Proof: 1) Show $\overline{\text{2SAT}} \in$ NL (good exercise)

2) Show $\text{PATH} \leq_m^L \overline{\text{2SAT}}$

Give a log-space reduction f from PATH to $\overline{\text{2SAT}}$.

$$f(\langle G, s, t\rangle) = \langle\varphi\rangle$$

- For each node u in G put a variable x_u in φ.
- For each edge (u, v) in G, put a clause $(x_u \rightarrow x_v)$ in φ [equivalent to $(\overline{x_u} \vee x_v)$].
- In addition, put the clauses $(x_s \vee x_s)$ and $(x_t \rightarrow \overline{x_s})$ in φ.

Show G has a path from s to t iff φ is unsatisfiable.

($\rightarrow$) If G has a path from s to t, follow the corresponding implications in φ to get a contradiction.

($\leftarrow$) If G has no path from s to t, then assign all x_u TRUE where u is reachable from s, and all other variables FALSE. That gives a satisfying assignment to φ.

Show f is computable in log-space: straightforward.

Quick review of today

1. Log space: L and NL
2. Configuration graph
3. L ⊆ NL ⊆ P
4. Log-space reduction
5. PATH is NL-complete
6. $\overline{\mathrm{2SAT}}$ is NL-complete

These slides are adapted from Michael Sipser, *18.404J Theory of Computation*, Fall 2020. Massachusetts Institute of Technology:

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